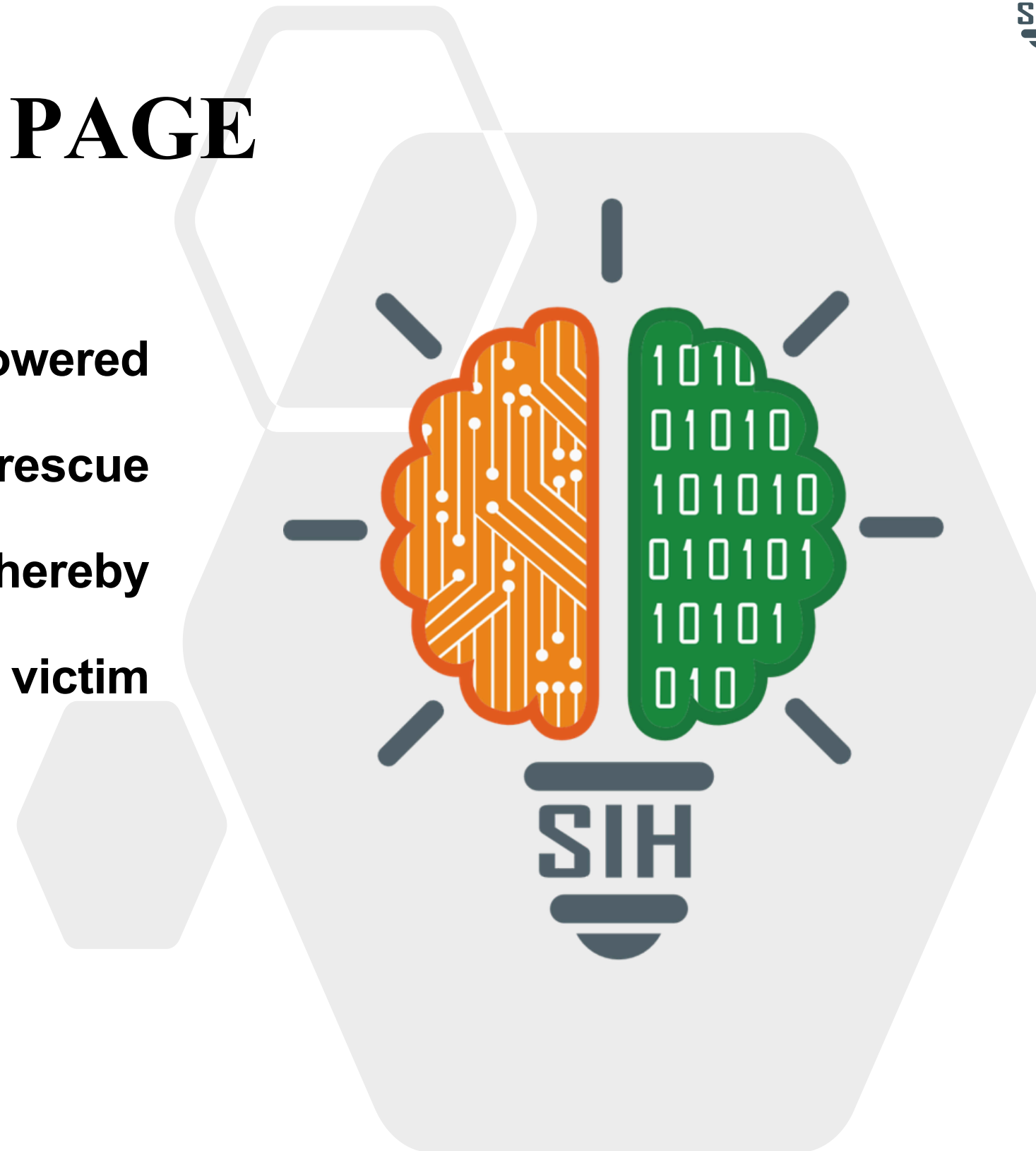


TITLE PAGE

- Problem Statement ID – SIH26177
- Problem Statement Title- A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.
- Theme- Robotics and Drones
- PS Category- Hardware
- Team ID- IIC_TMSL_P13_H21
- Team Name - SaResQ



SaResQ: AI for Disaster Response

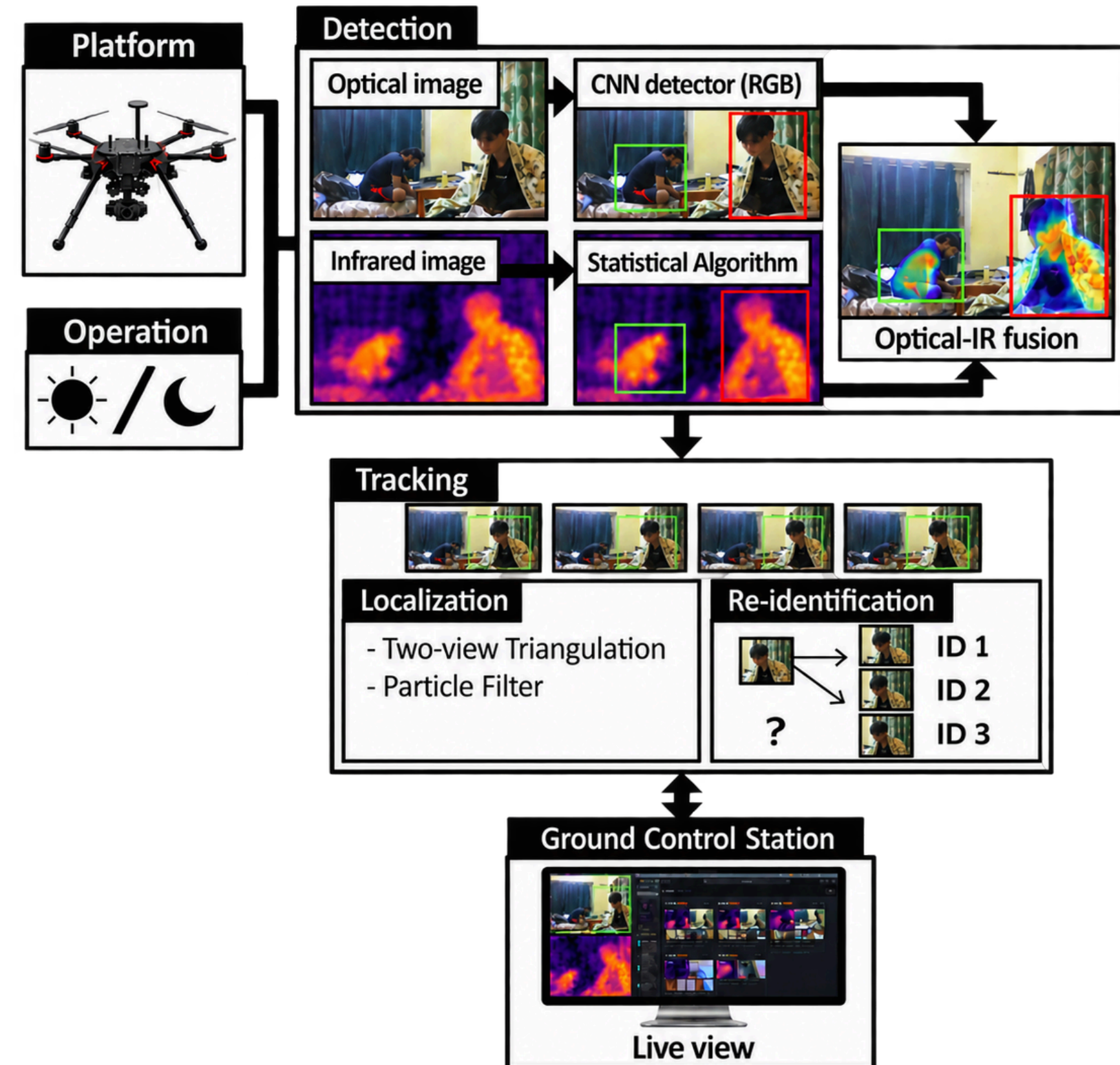
Detecting survivors and hazards using edge AI

PROBLEM

- **Limited visibility** — Survivors and hazards are difficult to locate after disasters.
- **Difficult terrain** — Damaged and inaccessible areas make ground search slow and risky.
- **Manual analysis** — Reviewing large volumes of aerial footage delays critical detections.

OUR SOLUTION

- **Thermal-first detection** — Identifies suspicious regions and detects survivors using lightweight YOLOv8n.
- **Disaster scene analysis** — Classifies fire, flooding, collapsed buildings, and other disaster conditions.
- **Evidence-based alerts** — Fuses multiple observations to improve confidence and geo-tag detected survivors.



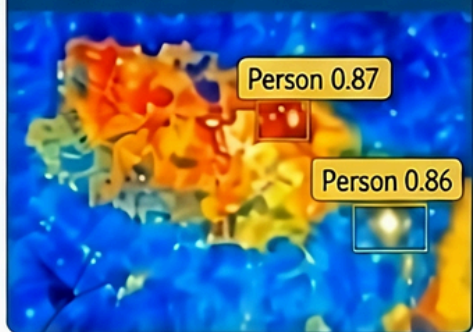
PROCESS FLOWCHART

SENSE

SENSED SYSTEM OUTPUT – RGB VIEW

RGB Frame
(1280 × 720)Thermal Frame
(32 × 24)

SENSED SYSTEM OUTPUT – THERMAL VIEW



FIND CANDIDATES

THERMAL CONTRAST ANALYSIS:

- Perform contrast-relative thermal analysis (not fixed absolute threshold)
- Compare local thermal values with estimated score-background temperature

If thermal
contrast is strong:GENERATE
THERMAL
PROPOSALSIf thermal
contrast is weak:ACTIVATE RGB
SALIENCY
FALLBACK

ADAPTIVE GATE:

Widens window when thermal contrast falls

ROI SELECTION:

Select 3–5 crops of 160 × 160 and
pass to the learned detector.

Supporting Hazard-Analysis Branch

HAZARD ANALYSIS

MobileNetV2: analyses surrounding
scene for environmental hazards.

IDENTIFY

RGB-THERMAL REGISTRATION:

Align thermal candidate region with
corresponding RGB region

YOLOv8n + P2 DETECTOR:

Run only on selected crops (INT8 inference)

Bounding box +
detection confidence

EVIDENCE ASSEMBLY

- RGB detection confidence
- RGB bounding-box area
- Thermal blob area
- RGB-thermal contrast
- Thermal blob eccentricity
- RGB-thermal IoU
- Ambient temperature
- Time of day
- Tracking information

Contextual
Info

LEARNED CROSS-MODAL FUSION

Pass combined features into a
small learned fusion classifierSurvivor
probability

CONFIRM

HIGH

RGB detection and thermal signature are
co-registered and mutually consistent.

Action: SURVIVOR LOCATED → DISPATCH

MEDIUM

Thermal signature is present but there is no
corresponding visible detection.Action: POSSIBLE OBSCURED SURVIVOR →
INVESTIGATE

LOW

Visible detection exists without thermal
corroboration.Action: POSSIBLE FALSE POSITIVE →
DEPRIORITISE

LOCATE & REPORT

FINAL OUTPUT: Geo-tagged survivor/hazard
information for the rescue team

Technologies Used

Hardware & Sensors

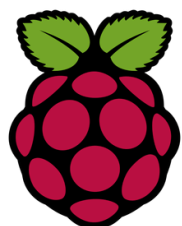
- Raspberry Pi 4
- MLX90640 Thermal Camera
- Pi Camera v2 (RGB)
- NEO-6M GNSS
- MPU-6050 IMU

AI / COMPUTER VISION

- YOLOv8n + P2
- MobileNetV2
- RGB–Thermal Fusion
- INT8/TFLite

SOFTWARE

- Python
- Raspberry Pi OS
- SQLite
- Flask
- Leaflet



Flask



TECHNICAL FEASIBILITY

- Built on a validated reference (99.7% CNN accuracy, peer-reviewed)
- Cascade design lets a CNN run on cheap, GPU-less hardware
- Fixes a known gap: thermal detection failing above 30°C
- Flight control uses proven open-source stack, not custom-built

FINANCIAL FEASIBILITY

- AI/SAR system BOM: ₹13,000
- Estimated complete autonomous drone: ~₹30,000
- ~7–10× cheaper than current thermal/SAR-capable enterprise drones (typically ~₹3–22 lakh)
- ~300–350g payload — compatible with commodity multirotor platforms (targeted 250g with lightweight power banks)
- ₹0 training compute cost — model trained using Google Colab

OPERATIONAL FEASIBILITY

- Airframe-agnostic — deploys on any F450-class commodity drone
- Fully offline by design — detection, classification, and geo-tagging need no connectivity
- Characterized from a fixed platform first, then flight-validated — isolates results from vibration/wind confounds

POTENTIAL CHALLENGES & RISK MITIGATION

THERMAL SENSOR DELAY

● **Likelihood:** Medium • **Impact:** HIGH

MITIGATION

Order early + dataset backup

CALIBRATION DRIFT

● **Likelihood:** Medium • **Impact:** HIGH

MITIGATION

Pre-flight re-check

LOW INFERENCE FPS

● **Likelihood:** Medium • **Impact:** HIGH

MITIGATION

Quantize + optimize

NO AIRFRAME FOR VALIDATION

● **Likelihood:** Medium • **Impact:** LOW

MITIGATION

Ground validation first

LIMITED TEMPERATURE TRIALS

● **Likelihood:** Medium • **Impact:** HIGH

MITIGATION

Day/night testing

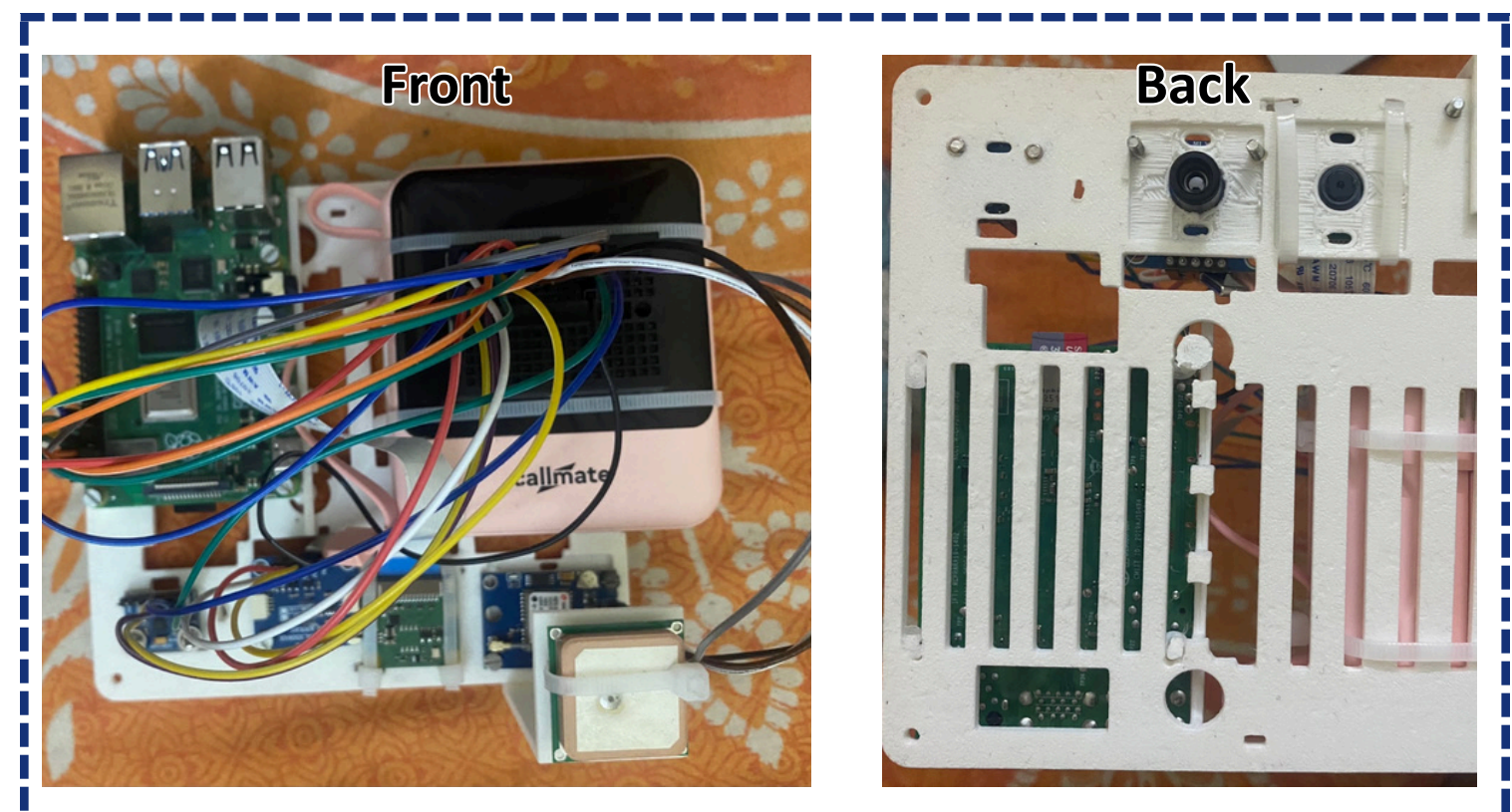
COMPUTE BOARD DAMAGE

● **Likelihood:** Low • **Impact:** HIGH

MITIGATION

Voltage checks + backup

PROTOTYPE VALIDATION



WHO BENEFITS

01 | Rescue Teams

Faster survivor identification and location.

02 | Disaster Response Agencies

Better aerial awareness during large-area searches.

03 | Survivors

Higher chance of detection at night and in low visibility.

04 | Drone Operators

Less manual monitoring with AI-assisted detection.

WHERE THE VALUE SHOWS UP

01 | Faster Search

AI prioritises areas where survivors may be present.

02 | Day & Night Detection

Thermal imaging supports detection in low visibility.

03 | Accurate Location Sharing

Geo-tagged detections guide rescue teams to the target.

04 | Low-Cost Deployment

Raspberry Pi 4 enables onboard AI processing.

WHY IS IT DIFFERENT?

- **Designed for India:** Adaptive thermal detection for reliable operation in high-temperature environments.
- **Lower Latency:** Edge AI processing enables faster detection and response.
- **Cost-Effective:** Core AI detection system developed at ~₹13K.

01 — UAV Human Detection

Deep Learning-based Human Detection for UAVs with Optical and Infrared Cameras: System and Experiments

T. Hinzmann et al. — ETH Zurich [[View PDF ↗](#)]

Used for: RGB + LWIR detection, tracking, localization & re-identification framework.

02 — RGB + Thermal Detection

Using Deep Learning and Low-Cost RGB and Thermal Cameras to Detect Pedestrians in Aerial Images Captured by Multirotor UAV

D. C. de Oliveira & M. A. Wehrmeister — 2018 [[View PDF ↗](#)]

Used for: Low-cost UAV vision and CNN-based pedestrian detection.

03 — Multimodal Fusion

Transformer Fusion-Based Scale-Aware Attention Network for Multispectral Victim Detection

Y. Chen, Y. Li, W. Zheng & X. Wan — 2024 [[View PDF ↗](#)]

Used for: RGB–thermal fusion and detection of victims at different scales.

04 — Tiny Object Detection

COXNet: Cross-Layer Fusion with Adaptive Alignment and Scale Integration for RGBT Tiny Object Detection

P. Peng et al. [[View PDF ↗](#)]

Used for: Cross-modal fusion, spatial alignment and tiny-object detection in drone imagery.

Research Gap: RGB + thermal detection works well, but small targets and real-time detection on low-cost hardware remain challenging.